

TIEHO LARA

Geomatics & GIS Professional · Health Systems Analyst · Land Surveyor

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PROFILE

Geomatics professional with over a decade of professional practice, five years of it in geomatics and GIS, integrating classical land surveying with spatial analytics, remote sensing and health-systems analysis. I design and run the full pipeline — field survey and UAV photogrammetry, spatial database design, WebGIS deployment, and decision-support reporting for ministries and donors. Current work spans health systems strengthening for Lesotho's Ministry of Health, national health catchment delineation and digital microplanning, national spatial data infrastructure and geospatial standards, and private-sector land survey consultancy. BSc Geomatics (Land Surveying), University of Cape Town; National Diploma, Civil Engineering Technology.

CORE COMPETENCIES

Surveying & geodesy — control networks, projections and datums, spatial reference frameworks, setting out, as-built verification

GIS & spatial analysis — catchment delineation, accessibility and equity analysis, spatial data validation, metadata and quality assurance

Remote sensing — UAV photogrammetry, aerial imagery processing, DSM/DTM and orthomosaic production

Health systems & M&E — referral analysis, HRH and resource mapping, immunisation coverage and equity, DHIS2, IDSR/eIDSR, MEAL frameworks

Standards & governance — ISO 19111/19115/19157, OGC services, AFREF, ICAO Annex 4/15, NSDI policy, WHO RED microplanning

Engineering — structural design and detailing, inspections, regulatory compliance, SANS-guided practice

PROFESSIONAL EXPERIENCE

Health Systems Strengthening Analyst (Volunteer) — Clinton Health Access Initiative, supporting the Ministry of Health

Sep 2026 – Nov 2026

- Analysing specialist referrals from Queen 'Mamohato Memorial Hospital to Bloemfontein — referral volumes, conditions, specialties, expenditure, reasons and patient pathways — and supporting their classification by clinical appropriateness and local manageability.
- Documenting the workforce, skills, equipment, diagnostics, infrastructure and commodity constraints driving out-of-country referral, and contributing to a phased referral optimisation and investment plan.
- Human resources for health optimisation: analysis of approved establishments, filled and vacant posts, geographic and facility-level distribution, skills mix, workload and externally funded cadres, identifying staffing gaps, maldistribution and task-sharing options.
- National resource mapping and expenditure tracking: partner follow-up, submission review, reconciliation and consolidation across funding sources, programme areas and years, supporting the GC8 and GAVI 6.0 applications.
- Producing analysis, dashboards, decks and briefs for Ministry leadership and technical teams; designing confidential data-collection instruments in KoboToolbox and maintaining decision and action records.

Reports to the Senior Technical Advisor, Health Systems Strengthening. All outputs remain the property of CHAI and/or the Ministry of Health and are described here at scope level only.

Senior GIS Consultant — GDI Advisors (Ministry of Health EPI programme, Gavi-funded)

Apr 2025 – Sep 2026

- Led national and district geospatial pre-analysis for health catchment delineation across all ten districts of Lesotho, establishing the first complete national GIS dataset for catchment analysis — approximately 145 health facilities, 6,225 villages and 2,793 schools.
- Delivered a village reassignment analysis covering 178 villages, reducing total village-to-facility travel distance by 2,908 km — a 47% reduction, averaging 16.3 km per village reassigned.
- Designed and implemented end-to-end GIS data pipelines: data structuring, validation, metadata assignment and quality assurance aligned to national systems and NSDI foundations.
- Lead author of the 82-page donor closeout report with district-level synthesis, methodology, nine recommendations and alignment to the WHO/UNICEF/Gavi geo-enabled microplanning blueprint, plus seven district and technical reports.
- Built and published an interactive WebGIS prototype of facility catchments and village assignments, and specified interoperability with Ministry of Health systems.

- Designed and facilitated the national capacity-building programme — a central Training of Trainers and district training with facilitator guide, participant workbook and session plans; participants rated district relevance 4.9/5.
- Drafted terms of reference for a Ministry-hosted EPI GIS Specialist to institutionalise the programme beyond consultancy.

Founding Director & Land Surveyor — Giri Group (Pty) Ltd t/a Giri Consultants

May 2022 – Present

- Direct a survey, GIS and field data collection practice serving private, institutional and government clients from Thaba Bosiu, Maseru.
- Deliver topographic, photogrammetric and engineering surveys; establish and verify control networks, benchmarks and reference points; produce orthomosaics, survey drawings and as-built records using GNSS, total station and levelling instruments, with drone imagery processed against surveyed ground control.
- Survey lead on a large mixed-use development (filling station, shopping facilities, entertainment venues), authoring the integrated survey, design and approvals workflow — lease structuring under the Land Act 2010, subdivision and servitude strategy, project spatial reference framework, and the statutory critical path from allocation through to occupancy certificate.
- Delivered ArcGIS Online licence supply and implementation to the Ministry of Health (2026): named-user allocation, installation and configuration, system commissioning and formal user acceptance.
- Environmental sensitivity mapping and field verification, Polihali, Mokhotlong (2023).

Cadastral diagrams requiring lodgement are prepared for sign-off by a registered land surveyor; Lesotho's Land Surveyors' Board is not currently issuing licences.

Geospatial Standards & NSDI Technical Consultant — GIZ, for the Ministry of Local Government, Chieftainship, Home Affairs and Police

Feb 2026 – Mar 2026

- Assessed Lesotho's National Spatial Data Infrastructure against ISO, OGC, ICAO and AFREF standards, covering data governance, UAV governance and coordinate reference system practice.
- Developed standards-aligned assessment methodologies and scoring tools, and produced donor-ready reporting with prioritised policy and technical recommendations.

Analyst, GIS & Immunisation M&E — Clinton Health Access Initiative (CHAI)

Aug 2023 – Apr 2025

- Provided spatial intelligence to the Ministry of Health Expanded Programme on Immunisation: coverage, equity and access analysis across all ten districts, with spatial consistency maintained across administrative levels.
- Built interactive dashboards linked to DHIS2 for district-level equity analysis, dropout tracking and coverage monitoring, and supported the HPV vaccination campaign with targeted spatial analysis of under-served areas.
- Designed M&E frameworks for health-centre data clerks and trained them on GIS and M&E tools; conducted routine data audits and standardised validation against WHO and national health data standards.
- Supported IDSR / eIDSR surveillance reporting and aligned GIS outputs with HMIS and national data standards.

GIS Analyst — Remote Sensing & Disaster Risk Reduction — World Vision Lesotho

Jan 2022 – Aug 2023

- Led aerial photogrammetry and UAV-based mapping for disaster risk reduction, managing three drone pilots and processing more than 50 photogrammetry datasets into hazard and flood modelling maps.
- Co-developed the national MHEWED multi-hazard early-warning web system with Lesotho Meteorological Services, improving early warning and early detection.
- Produced mapping and spatial analysis supporting Community Disaster Preparedness Plans across four districts, and GIS-driven reporting that informed the ECHO Phase IV funding case.

Civil & Structural Engineering Consultant — 10 Talents Design Build

Aug 2015 – Dec 2021

- Structural design and detailing in concrete, steel and timber; engineering surveying and CAD design on infrastructure and building projects; more than ten completed projects.
- Site inspections and regulatory compliance under the Building Control Act 1995 and the Town and Country Planning Act 1980, to SANS-guided practice.

Intern, Field Survey & Data Processing — Cape Survey, Cape Town

Jun 2018 – Jul 2018

Intern, Structural Section — Ministry of Public Works and Transport

Jun 2013 – Jun 2015

- Building inspections, CAD documentation, structural modelling, design loads and planning approval support.

PROFESSIONAL LEADERSHIP

Lesotho Geomatics Institute (LGI) transition — Lesotho Land Surveyors Association

2026 – Present

- Leading the revival of the Association as the Lesotho Geomatics Institute, a geospatial-science platform modelled on the SAGI / SAGC pair, working to the Land Survey Act 1980 as amended by Act No. 2 of 2019.

- Constitutional review, formal amendments and committee role definition; a member categorisation framework across Land Surveying, GISc and Mine Surveying branches at professional, technologist and technician tiers with candidacy and articles; and the association's web platform.

EDUCATION & PROFESSIONAL TRAINING

BSc Geomatics Engineering (Land Surveying) — University of Cape Town, South Africa · 2018 – 2021

National Diploma, Civil Engineering Technology — Lerotholi Polytechnic, Lesotho · 2012 – 2015

Postgraduate Diploma, Monitoring, Evaluation, Accountability & Learning — Humanitarian Global Institute
· from 2024, in progress

Certificates — ArcGIS Pro Essential Training; BIM Foundations; Learning Tekla BIMsight; Learning Access 2016; Artificial Intelligence for Project Managers

Membership — Lesotho Association of Engineers (LAE)

TECHNICAL SKILLS

GIS & surveying — ArcGIS Pro, ArcGIS Online, QGIS, WebODM, AutoCAD, Civil 3D, Revit, Leaflet

Analytics — Python, Stata, Power BI, Tableau, advanced Excel

Health data — DHIS2, KoboToolbox, ODK, SurveyCTO

Structural — Prokon, Robot Structural Analysis, ETABS, SAP2000, SAFE, Tekla Structures, Tekla Structural Designer

Field instruments — GNSS receivers, total station, digital level, multirotor UAV

Project & collaboration — Asta Powerproject, Microsoft Project, Microsoft 365, Confluence, Notion

REFERENCES

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